FortiGate VM → Python Syslog Setup

Steps and commands to forward FortiGate logs to a local Python listener

1. Environment Overview

- Host PC IP (Ethernet): 192.168.1.2
- FortiGate VM IP (port1): 192.168.1.101
- VMware Network Mode: Bridged (manually set to Intel Ethernet adapter)
- Syslog Protocol: UDP on port 514

2. Python Syslog Listener

Make sure your Python backend is listening on UDP 0.0.0.0:514 before applying FortiGate config. Example minimal listener:

```
import socket

HOST = '0.0.0.0'
PORT = 514

sock = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)  # UDP
sock.bind((HOST, PORT))
print(f'Listening for syslog on UDP {PORT}...')

while True:
    data, addr = sock.recvfrom(4096)
    print(f'[{addr[0]}] {data.decode("utf-8", errors="replace").strip()}')
```

⚠ *On Windows, port 514 requires running the terminal as Administrator.*

3. Windows Firewall Rules

Open an elevated PowerShell and run both commands to allow inbound syslog and ICMP:

```
# Allow UDP 514 (syslog)
New-NetFirewallRule -DisplayName "Syslog UDP 514" -Direction Inbound -Protocol UDP
-LocalPort 514 -Action Allow

# Allow ICMPv4 (ping) — needed to verify connectivity
New-NetFirewallRule -DisplayName "Allow ICMP" -Protocol ICMPv4 -IcmpType 8 -
Direction Inbound -Action Allow
```

4. VMware Bridged Adapter Fix

The default 'Bridged (Automatic)' mode may pick the wrong adapter. Force it to your physical Ethernet:

1. In VMware Workstation: Edit → Virtual Network Editor
2. Click Change Settings (requires admin)
3. Select VMnet0 → change 'Bridged to:' from Automatic to your Intel Ethernet adapter
4. Click Apply → OK

> ⚠ *Only Network Adapter 1 should have 'Connected' and 'Connect at power on' checked. All others should be unchecked.*

5. FortiGate CLI — Syslog Configuration

SSH or console into FortiGate and run the following commands:

5.1 Configure Syslog Destination

```
config log syslogd setting
    set status enable
    set server "192.168.1.2"
    set port 514
    set mode udp
end
```

5.2 Configure Log Filter (severity)

```
config log syslogd filter
    set severity information
end
```

> ⚠ *The 'severity' option goes in 'log syslogd filter', NOT in 'log syslogd setting'. Mixing them causes a parse error.*

6. Verify Connectivity

After applying the config, confirm bidirectional connectivity:

6.1 From FortiGate — ping the host PC

```
execute ping 192.168.1.2

# Expected output:
# 64 bytes from 192.168.1.2: icmp_seq=0 ttl=128 time=0.2 ms
# 5 packets transmitted, 5 packets received, 0% packet loss
```

6.2 From Host PC — ping the FortiGate

```
ping 192.168.1.101

# Expected output:
# Reply from 192.168.1.101: bytes=32 time<1ms TTL=255
```

7. Test Log Generation

Trigger test log events from FortiGate CLI to confirm the pipeline is working:

```
execute log test

# This generates sample events of all types:
# traffic, utm, virus, IPS, botnet, antispam, etc.
```

> ⚠ *'execute log test' may be unavailable on unlicensed FortiGate VM editions. Real traffic logs will still flow once the syslog config is applied.*

8. Expected Log Format

Logs arrive as standard syslog messages. Example of a traffic log received by the Python listener:

```
[syslog DEBUG] from 192.168.1.101: <189>date=2026-03-15 time=07:48:21
  devname="FortiGate-VM64" devid="FGVMSLTM26010201"
  type="traffic" subtype="local" level="notice" vd="root"
  srcip=192.168.1.101 srcport=7188 dstip=96.45.46.46 dstport=853
  proto=6 action="timeout" duration=12 sentbyte=180 rcvdbyte=0
```

9. Common Errors & Fixes

Error	Fix
`Command parse error before 'severity'`	Put 'set severity' in 'config log syslogd filter', not in 'config log syslogd setting'
`100% packet loss on ping`	Set VMnet0 in Virtual Network Editor to your specific Intel Ethernet adapter, not Automatic
`No logs in Python listener`	Ensure Python is running as Administrator and Windows Firewall rule for UDP 514 is added
`'execute log test' fails`	Normal on unlicensed VM — real traffic logs will still forward automatically
`Bridged VM gets wrong subnet`	Check Virtual Network Editor and confirm VMnet0 bridges to the 192.168.1.x Ethernet NIC